

# Randomized, Placebo-Controlled Trial of Mipomersen in Patients with Severe Hypercholesterolemia Receiving Maximally Tolerated Lipid-Lowering Therapy

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## Abstract

**Objectives:** Mipomersen, an antisense oligonucleotide targeting apolipoprotein B synthesis, significantly reduces LDL-C and other atherogenic lipoproteins in familial hypercholesterolemia when added to ongoing maximally tolerated lipid-lowering therapy. Safety and efficacy of mipomersen in patients with severe hypercholesterolemia was evaluated.

**Methods and Results:** Randomized, double-blind, placebo-controlled, multicenter trial. Patients (n = 58) were  $\geq 18$  years with LDL-C  $\geq 7.8$  mmol/L or LDL-C  $\geq 5.1$  mmol/L plus CHD disease, on maximally tolerated lipid-lowering therapy that excluded apheresis. Weekly subcutaneous injections of mipomersen 200 mg (n = 39) or placebo (n = 19) were added to lipid-lowering therapy for 26 weeks. Main outcome: percent reduction in LDL-C from baseline to 2 weeks after the last dose of treatment. Mipomersen (n = 27) reduced LDL-C by 36%, from a baseline of 7.2 mmol/L, for a mean absolute reduction of 2.6 mmol/L. Conversely, mean LDL-C increased 13% in placebo (n = 18) from a baseline of 6.5 mmol/L (mipomersen vs placebo  $p < 0.001$ ). Mipomersen produced statistically significant ( $p < 0.001$ ) reductions in apolipoprotein B and lipoprotein(a), with no change in high-density lipoprotein cholesterol. Mild-to-moderate injection site reactions were the most frequently reported adverse events with mipomersen. Mild-to-moderate flu-like symptoms were reported more often with mipomersen. Alanine transaminase increase, aspartate transaminase increase, and hepatic steatosis occurred in 21%, 13% and 13% of mipomersen treated patients, respectively. Adverse events by category for the placebo and mipomersen groups respectively were: total adverse events, 16(84.2%), 39(100%); serious adverse events, 0(0%), 6(15.4%); discontinuations due to adverse events, 1(5.3%), 8(20.5%) and cardiac adverse events, 1(5.3%), 5(12.8%).

**Conclusion:** Mipomersen significantly reduced LDL-C, apolipoprotein B, total cholesterol and non-HDL-cholesterol, and lipoprotein(a). Mounting evidence suggests it may be a potential pharmacologic option for lowering LDL-C in patients with severe hypercholesterolemia not adequately controlled using existing therapies. Future studies will explore alternative dosing schedules aimed at minimizing side effects.

**Trial Registration:** ClinicalTrials.gov NCT00794664.

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## Introduction

Familial hypercholesterolemia (FH), an autosomal dominant disorder characterized by elevated plasma levels of low-density lipoprotein cholesterol (LDL-C) is most commonly caused by a

mutation in the LDL receptor gene. [1] The risk of a coronary event before age 65 is 85% in men and 50% in women with untreated heterozygous FH. [2] Although high dose statin therapy in combination with other lipid-lowering agents is effective in these

patients, a small percentage have refractory severe hypercholesterolemia.[3–6] In small non-randomized studies LDL-apheresis reduces cardiac events and improves overall survival; however, it has limited availability, generally requires a fistula, and costs over €2500 for biweekly sessions. [5,7] In the US, fewer than 200 of the 6000 qualified persons with severe hypercholesterolemia receive apheresis. [7] Agents capable of lowering LDL-C via novel mechanisms of action are needed.

In previous trials, weekly subcutaneous (SC) injections of mipomersen 200 mg reduced LDL-C by 25% and 28% in patients with FH when added to existing treatments. [8,9] Mipomersen is an antisense oligonucleotide inhibitor of apolipoprotein B (Apo B) synthesis. It binds to messenger RNA encoding apolipoprotein B preventing its synthesis and secretion of Apo B containing atherogenic lipoproteins. [8,10] This study assessed the safety and efficacy of mipomersen 200 mg per week in patients with severe hypercholesterolemia receiving maximally tolerated lipid-lowering therapy that excluded apheresis and found treatment to significantly lower atherogenic lipoproteins.

## Methods

### Ethics Statement

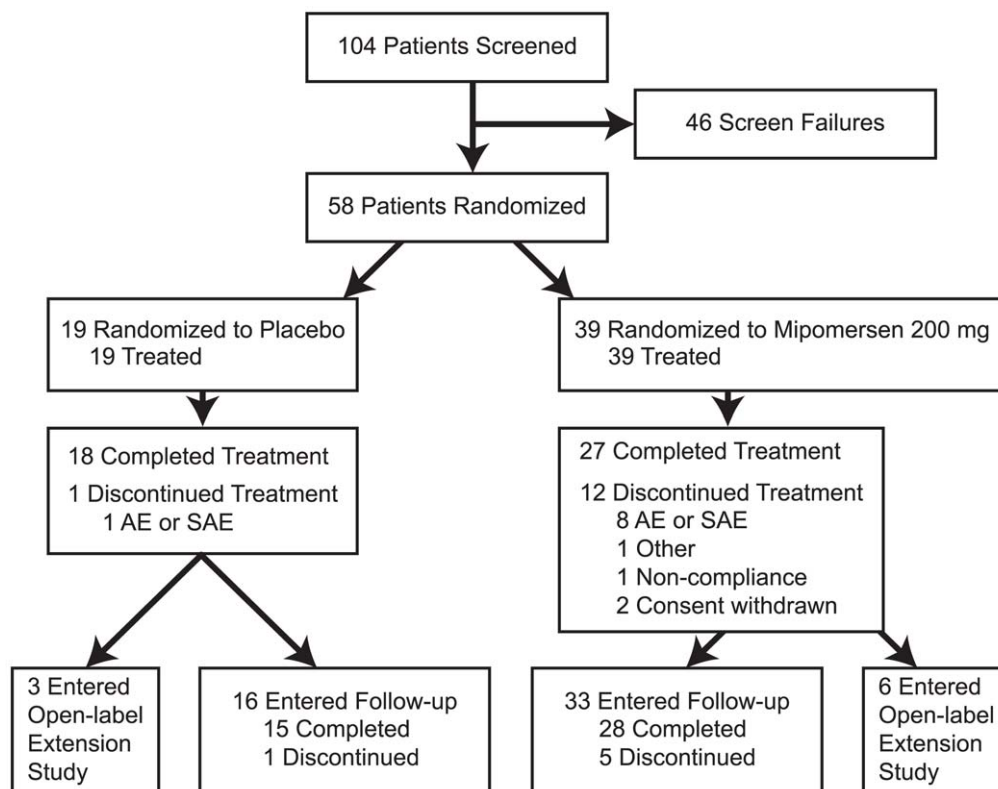
This randomized, double-blind, placebo-controlled, multi-center, Phase 3 trial was undertaken between January 27, 2009 and October 14, 2010 and performed in accordance with the Declaration of Helsinki with approval by each Institutional Review Board. This multicenter study included 26 study locations; the names of Institutions and their corresponding institutional review boards/independent ethics committees are listed in the supplemental material. At screening and after adequate explanation of the aims, methods, anticipated benefits, and potential

hazards of the study, all patients provided written informed consent.

### Study Design

The protocol for this trial and supporting CONSORT checklist are available as supporting information; see Checklist S1 and Protocol S1. Adults patients with severe hypercholesterolemia defined as an LDL-C  $\geq 5.1$  mmol/L with known CHD or an LDL-C  $\geq 7.8$  mmol/L in the absence of known CHD provided written informed consent. [11] Patients were on a stable low fat diet, at a stable weight, on maximally tolerated lipid-lowering therapy, met LDL-apheresis criteria but apheresis was prohibited.[12–14] Exclusions included significant cardiovascular or cerebrovascular events within 24 weeks of screening, congestive heart failure, type I diabetes, poorly controlled type II diabetes, hypertension, secondary hyperlipidemia predisposition, or a history of significant renal or hepatic disease.

After 4-weeks screening, eligible patients were randomized to 26 weeks of self-administered mipomersen 200 mg per week or placebo, which consisted of 9 mg of sodium chloride, 0.004 mg of riboflavin, filled to sufficient quantity with 1 mL water for injection. The placebo was similar to active drug in appearance and administration. Permuted block randomization (blocks of size 3) was used to allocate patients to treatment in a 2:1 ratio. There were no stratification factors used; 36 sites screened patients and 26 sites enrolled and randomized at least 1 patient. Patients were enrolled by the principal investigator at each study site after which the investigator or study coordinator called an interactive voice response system that provided blinded medication kits coded to assure that all clinical, medical, and pharmacy personnel, as well as the patient, remained blinded to treatment allocation. All



**Figure 1. Patient Disposition – Consort Diagram.**

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**Table 1.** Baseline Characteristics.

|                                                                              | Placebo      | Mipomersen   | Total        |
|------------------------------------------------------------------------------|--------------|--------------|--------------|
|                                                                              | (N = 19)     | (N = 39)     | (N = 58)     |
| Age (years) Mean (SD)                                                        | 47.9 (13.5)  | 51.8 (14.3)  | 50.5 (14.0)  |
| [Min, Max]                                                                   | [18, 66]     | [21, 77]     | [18, 77]     |
| Gender n (%)                                                                 |              |              |              |
| Male                                                                         | 7 (36.8)     | 18 (46.2)    | 25 (43.1)    |
| Female                                                                       | 12 (63.2)    | 21 (53.8)    | 33 (56.9)    |
| Race n (%)                                                                   |              |              |              |
| White                                                                        | 16 (84.2)    | 33 (84.6)    | 49 (84.5)    |
| Black                                                                        | 1 (5.3)      | 2 (5.1)      | 3 (5.2)      |
| Asian                                                                        | 0 (0.0)      | 1 (2.6)      | 1 (1.7)      |
| Multiple                                                                     | 2 (10.5)     | 1 (2.6)      | 3 (5.2)      |
| Other                                                                        | 0 (0.0)      | 2 (5.1)      | 2 (3.4)      |
| BMI (kg/m <sup>2</sup> )                                                     |              |              |              |
| Mean (SD)                                                                    | 29.9 (4.10)  | 28.4 (5.35)  | 28.9 (4.98)  |
| [Min, Max]                                                                   | [22.8, 36.9] | [19.4, 39.8] | [19.4, 39.8] |
| Waist/hip ratio                                                              |              |              |              |
| Mean (SD)                                                                    | 0.94 (0.06)  | 0.92 (0.09)  | 0.93 (0.08)  |
| [Min, Max]                                                                   | [0.8, 1.0]   | [0.7, 1.1]   | [0.7, 1.1]   |
| Current smoker n (%)                                                         | 5 (26.3)     | 4 (10.3)     | 9 (15.5)     |
| Alcohol use n (%)                                                            |              |              |              |
| Current                                                                      | 7 (36.8)     | 27 (69.2)    | 34 (58.6)    |
| Non-current                                                                  | 4 (21.1)     | 5 (12.8)     | 9 (15.5)     |
| Never                                                                        | 8 (42.1)     | 7 (17.9)     | 15 (25.9)    |
| Metabolic syndrome, n (%) <sup>a</sup>                                       |              |              |              |
| No                                                                           | 10 (52.6)    | 25 (64.1)    | 35 (60.3)    |
| Yes                                                                          | 9 (47.4)     | 14 (35.9)    | 23 (39.7)    |
| History of:                                                                  |              |              |              |
| Angina                                                                       | 5 (26.3)     | 11 (28.2)    | 16 (27.6)    |
| Myocardial infarction                                                        | 4 (21.1)     | 8 (20.5)     | 12 (20.7)    |
| Coronary Artery Bypass Graft                                                 | 6 (31.6)     | 12 (30.8)    | 18 (31.0)    |
| Percutaneous Coronary Intervention                                           | 2 (10.5)     | 4 (10.3)     | 6 (10.3)     |
| Coronary Artery Disease without event                                        | 5 (26.3)     | 11 (28.2)    | 16 (27.6)    |
| Peripheral Artery Disease                                                    | 2 (10.5)     | 1 (2.6)      | 3 (5.2)      |
| Abdominal Aortic Aneurysm                                                    | 0 (0.0)      | 1 (2.6)      | 1 (1.7)      |
| Carotid Disease                                                              | 5 (26.3)     | 6 (15.4)     | 11 (19.0)    |
| Coronary Heart Disease or other Atherosclerotic disease (clinical diagnosis) | 15 (78.9)    | 31 (79.5)    | 46 (79.3)    |
| Lipid Lowering Medications, n (%)                                            | 19 (100.0)   | 39 (100.0)   | 58 (100.0)   |
| Maximal statin                                                               | 8 (42.1)     | 16 (41.0)    | 24 (41.4)    |
| Statin with ezetimibe                                                        | 11 (57.9)    | 20 (51.3)    | 31 (53.4)    |

Abbreviations: BMI, body mass index; Max, maximum; Min, minimum; SD, standard deviation.

<sup>a</sup>Metabolic syndrome determined according to the American Heart Association and the National Heart, Lung, and Blood Institute definition.

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clinical operations, data management, and statistical personnel also remained blinded until the database was locked. Patients underwent baseline liver and spleen magnetic resonance (MRI) or computed tomography with follow-up only in patients experiencing an ALT  $\geq 3$  X ULN on 2 consecutive results one week apart. MRI of hepatic fat content was performed by an assessor masked to treatment. [15].

Total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C) and triglycerides (TG) were measured by enzymatic colorimetry. Lipoprotein(a) [Lp(a)] was measured by a standardized isoform independent assay. Rate nephelometry assessed apolipoprotein B-100 and apolipoprotein A1. [16,17] By protocol, discontinuation was required if alanine transaminase (ALT) or aspartate aminotransferase (AST) levels were:  $\geq 8$  X ULN,  $\geq 5$  X ULN on 2 consecutive weeks, or  $\geq 3$  X ULN in conjunction with